

EE 3001 Telecommunication System Design with DSP Capsule

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Abstract—This paper presents a complete end-to-end wireless speech communication system that integrates a low-bit-rate Linear Predictive Coding (LPC) speech analyzer/synthesizer with a Binary Phase-Shift Keying (BPSK) physical layer realized on a Universal Software Radio Peripheral (USRP). A corrupted speech recording sampled at 48 kHz is conditioned by a zero-phase FIR low-pass filter and decimated to 8 kHz. Each 20 ms frame is modelled by a 10th-order LPC filter represented through Line Spectral Frequencies (LSFs) and augmented by gain, pitch, and a voiced/unvoiced flag, yielding a compact 97-bit/frame packet. The parameters are scalar-quantized, serialized, BPSK-modulated, and shaped by a square-root raised-cosine (RRC) filter. The waveform is evaluated over two channels: an ideal additive white Gaussian noise (AWGN) simulation and a real over-the-air USRP B200 link at 900 MHz. At the receiver, envelope cross-correlation synchronization, block-wise carrier-phase tracking, and 180° phase-ambiguity resolution recover the bits, after which an excitation-plus-all-pole synthesizer reconstructs intelligible speech. The source coder reduces the 128 kbit/s PCM stream to a 9.7 kbit/s parameter stream, a compression of about 13.2:1. The simulated link attains a bit-error rate (BER) of 0 at an effective SNR of 19.0 dB, whereas the hardware link operates at an estimated SNR of 0.7 dB with a measured BER of 2.7×10^{-2} , quantifying the gap between idealized and physical conditions.

Index Terms—Linear predictive coding, line spectral frequencies, speech coding, BPSK, software-defined radio, USRP, AWGN, bit-error rate, pulse shaping, vocoder.

I. INTRODUCTION

Transmitting raw speech waveforms over a wireless link is spectrally inefficient. A speech signal sampled at 8 kHz with 16-bit linear pulse-code modulation (PCM) already requires 128 kbit/s, even though the perceptually relevant information in voiced and unvoiced speech can be captured by a far smaller set of slowly varying parameters. Parametric coders exploit this redundancy by modelling the short-time spectral envelope produced by the human vocal tract together with a simple excitation description, an approach that underlies practically every modern digital-voice standard.

This work realizes the canonical analysis–transmission–synthesis pipeline of a vocoder as a physical communication system, as required by the EE 3001 capsule project. A corrupted 48 kHz recording is conditioned and decimated to 8 kHz, modelled frame-by-frame by 10th-order LPC expressed as LSFs, supplemented with gain, pitch and voicing, quantized to 97 bits per frame, and transmitted with BPSK. The link is exercised both in a

controlled AWGN simulation and physically over the air through a USRP B200 software-defined radio, and the recovered parameters are used to resynthesize speech at the receiver.

The contributions of the report are: (i) a robust front-end that removes out-of-band corruption and aliasing before downsampling; (ii) a frame-based LPC/LSF analyzer with autocorrelation pitch estimation and a three-feature voiced/unvoiced classifier; (iii) a fixed-rate bit-allocation scheme and the resulting 13.2:1 data-size reduction; (iv) a complete BPSK transmitter and receiver including RRC pulse shaping, matched filtering, synchronization, and carrier-phase recovery; and (v) a quantitative comparison of simulated versus real-hardware performance in terms of BER and SNR.

II. SYSTEM OVERVIEW

The system follows the block diagram of the project pipeline: a speech source feeds a DSP analysis stage; the resulting bitstream is BPSK-modulated and emitted by the USRP transmitter; it propagates through a channel (AWGN in simulation, the physical radio channel on hardware); the USRP receiver demodulates it; and a second DSP stage reconstructs the output speech. Two sampling-rate domains coexist. Speech analysis and synthesis operate in the 8 kHz speech domain, which is adequate for the 0–4 kHz telephone band, whereas the BPSK waveform is generated at a much higher rate — eight samples per symbol — so that pulse shaping, modulation, and the USRP front-end can operate cleanly.

Each processing block is verified by a dedicated diagnostic figure. The signal-preparation stage is documented by Figs. 1–5, the LPC/LSF analyzer by Figs. 6–11, the transmitter by Figs. 12–15, and the channel/receiver by Figs. 16–19. Aggregate performance is summarized in Figs. 20–22.

III. SIGNAL PREPARATION

A. Analysis of the Corrupted Input

The provided recording is sampled at 48 kHz and is visibly noisy in the time domain (Fig. 1). Its magnitude spectrum (Fig. 2) reveals the nature of the corruption: in addition to the genuine speech energy concentrated below 4 kHz, several strong narrowband components appear well above the 4 kHz speech band, extending toward 6–8 kHz. If the signal were decimated directly, these high-frequency components would alias into the speech band and irreparably contaminate the LPC model. The signal must therefore be band-limited before any rate reduction.

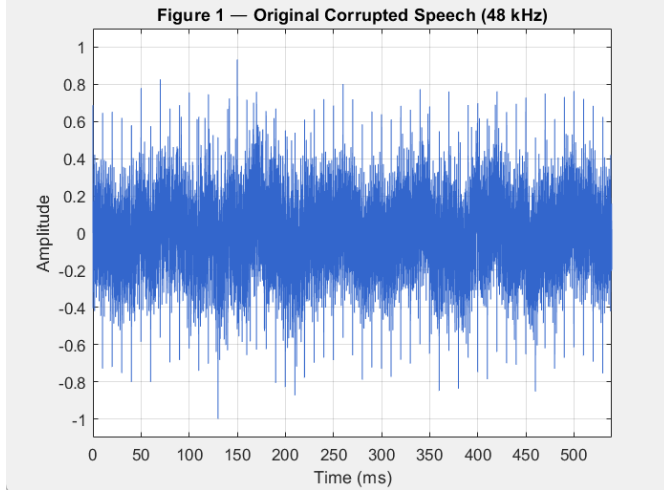


Fig. 1. Time-domain waveform of the corrupted 48 kHz input speech.

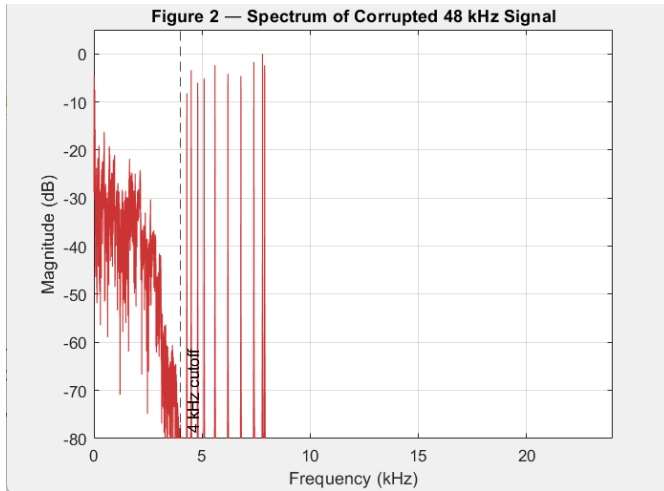


Fig. 2. Magnitude spectrum of the corrupted signal. Strong out-of-band components appear above the 4 kHz speech band.

B. Anti-Alias Low-Pass Filtering

A linear-phase FIR low-pass filter of order 120 is designed by the windowed-sinc method with a Hamming window. The cutoff is set to 3.6 kHz — deliberately below the 4 kHz Nyquist limit of the target 8 kHz rate — to provide a transition guard band. The normalized cutoff is

$$\omega_n = f_c / (f_s/2) = 3600 / 24000 = 0.15, \quad (1)$$

expressed relative to the Nyquist frequency. To avoid the group-delay distortion that an ordinary FIR filter would introduce, the filter is applied with zero-phase forward-backward filtering (*filtfilt*), which squares the magnitude response and cancels phase. Fig. 3 confirms that the out-of-band interferers are suppressed while the speech envelope is preserved.

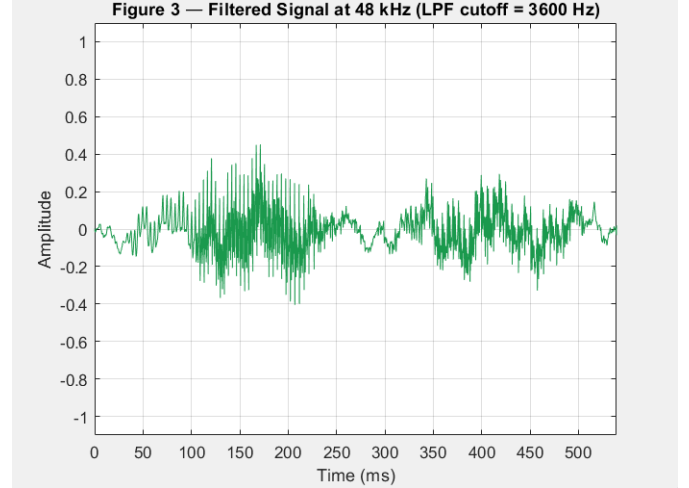


Fig. 3. Output of the zero-phase anti-alias filter (48 kHz, $f_c = 3.6$ kHz). Interference is removed.

C. Decimation to 8 kHz

The filtered signal is resampled by an integer factor of $M = 48000/8000 = 6$ using a polyphase implementation that incorporates its own anti-imaging filter. The resulting 8 kHz signal (Fig. 4) retains the speech structure. Fig. 5 overlays the 0–4 kHz spectra of the original 48 kHz signal and the processed 8 kHz signal: the speech band is preserved almost identically, while the spurious high-frequency content visible in Fig. 2 is absent. The signal is now suitable for the LPC/LSF stage.

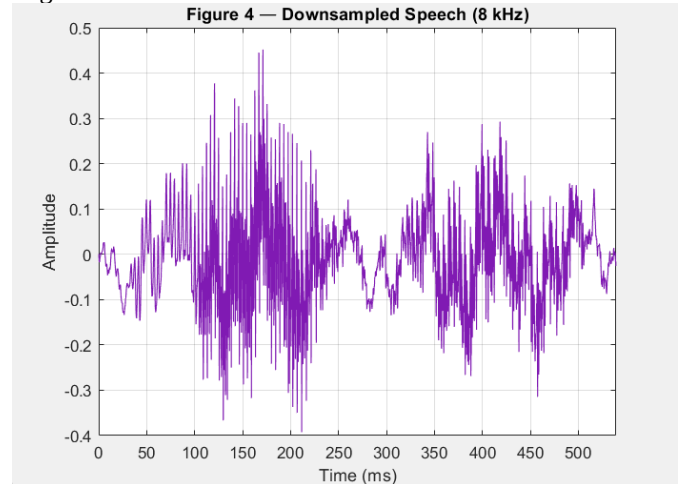


Fig. 4. Decimated speech at 8 kHz.

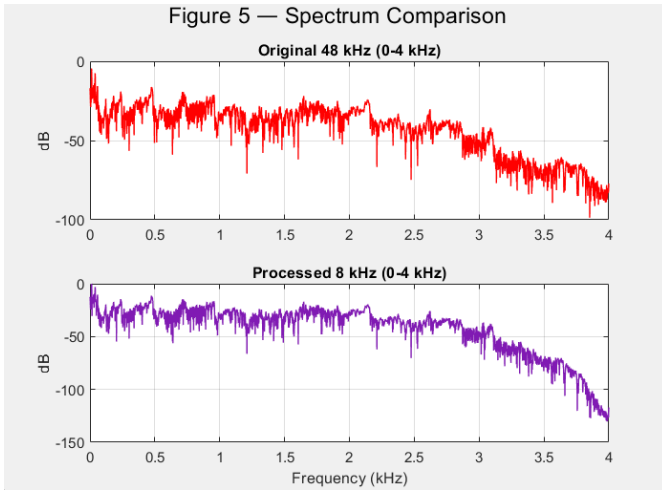


Fig. 5. Spectral comparison over 0–4 kHz: original 48 kHz (top) vs. processed 8 kHz (bottom).

IV. DSP-BASED SPEECH PARAMETER EXTRACTION

A. Framing and Windowing

Because speech is quasi-stationary only over short intervals, it is analyzed frame-by-frame. Each frame contains 160 samples (20 ms) and a 80-sample (10 ms) hop is used, giving 50% overlap and an analysis rate of about 100 frames per second. A Hamming window is applied to each frame before LPC analysis to reduce spectral leakage at the frame edges.

B. LPC Analysis and LSF Representation

Linear prediction models each speech sample as a weighted combination of its past values. For a p -th order predictor, the synthesis filter is the all-pole system

$$1/A(z) = 1/(1 + \sum_{k=1..p} a_k z^{-k}), \quad (2)$$

with the coefficients a_k obtained from the autocorrelation/Levinson–Durbin solution. An order of $p = 10$ is chosen, consistent with the usual rule of roughly one pole pair per kilohertz plus a margin, which is sufficient to capture the four to five formants present in an 8 kHz speech band.

Direct-form LPC coefficients are poorly suited to quantization and interpolation. They are therefore converted to Line Spectral Frequencies, defined from the symmetric and antisymmetric polynomials

$$P(z) = A(z) + z^{-(p+1)} A(z^{-1}), \quad (3)$$

$$Q(z) = A(z) - z^{-(p+1)} A(z^{-1}). \quad (4)$$

The LSFs are the angular positions of the roots of $P(z)$ and $Q(z)$ on the unit circle. They have two properties that make them ideal for transmission: they are ordered and bounded in $(0, \pi)$, and synthesis-filter stability is guaranteed as long as that ordering is preserved. This makes LSFs far more tolerant of quantization and bit errors than raw LPC coefficients. Fig. 6 shows the ten LSF trajectories over the utterance; they remain ordered and well separated and vary smoothly, tracking the moving formant structure of the speech.

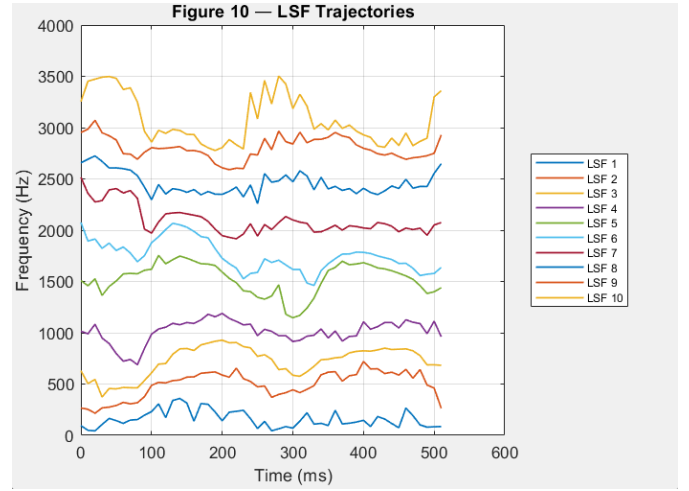


Fig. 6. Trajectories of the ten Line Spectral Frequencies across the utterance. The lines stay ordered and well separated, and follow the evolving formant structure.

C. Frame Energy and Gain

The short-time energy of frame m and the corresponding excitation gain are

$$E_m = \sum s^2[n], \quad g = \sqrt{(E_m / N)}. \quad (5)$$

The energy contour (Fig. 7) cleanly segments the utterance into high-energy voiced regions and low-energy silence/unvoiced regions, providing the first cue for the voicing decision.

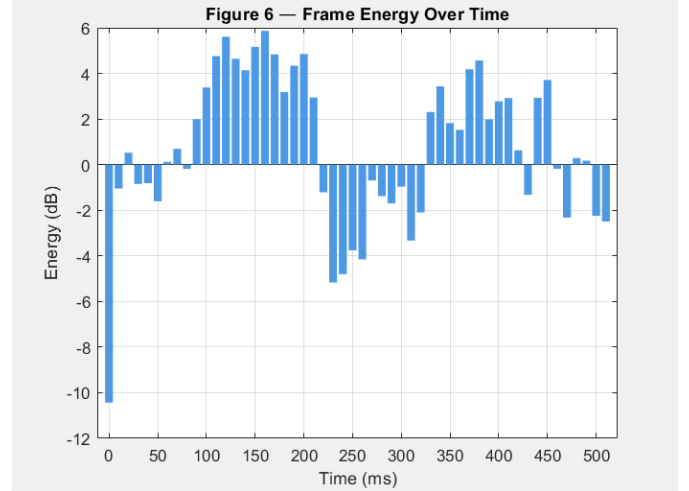


Fig. 7. Short-time frame energy (dB) over the utterance.

D. Pitch Estimation

Pitch is estimated per frame by autocorrelation. Each frame is first pre-emphasized by $1 - 0.95z^{-1}$ to flatten the spectral tilt, and the normalized autocorrelation is searched for its first strong peak within a lag range corresponding to 50–400 Hz, i.e. lags from $f_s/400 = 20$ to $f_s/50 = 160$ samples. A local-maximum constraint avoids selecting the zero-lag peak or spurious side lobes. The fundamental frequency is then $f_0 = f_s / \text{lag}$. Fig. 8 shows the extracted pitch contour, which lies in the expected 150–250 Hz range during voiced segments and is set to zero (marked) during unvoiced frames.

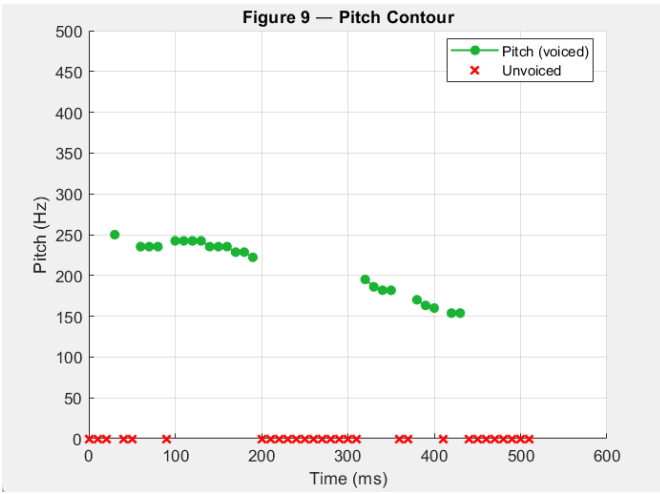


Fig. 8. Pitch contour. Voiced frames show 150–250 Hz; unvoiced frames are marked at zero.

E. Voiced/Unvoiced Decision

A robust three-feature classifier is used. A frame is declared voiced when at least two of the following hold and the autocorrelation test in particular is satisfied: energy above 30% of the median energy, normalized autocorrelation peak above 0.45, and zero-crossing rate (ZCR) below 0.18. The ZCR

$$ZCR = (1 / 2(N-1)) \sum |sign(s[n]) - sign(s[n-1])| \quad (6)$$

is high for noise-like unvoiced sounds and low for periodic voiced sounds. Fig. 9 plots the ZCR against its threshold, and Fig. 10 shows the resulting binary voiced/unvoiced decision, which agrees with the energy and pitch evidence. Combining three independent features by majority vote makes the classifier resilient to any single misleading cue.

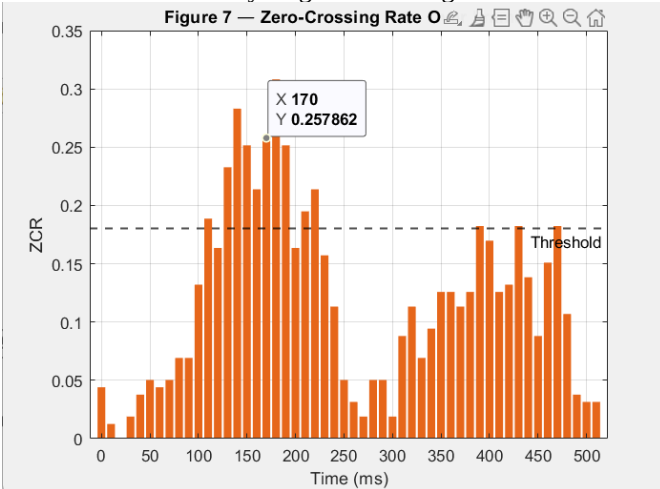


Fig. 9. Zero-crossing rate per frame with decision threshold (0.18).

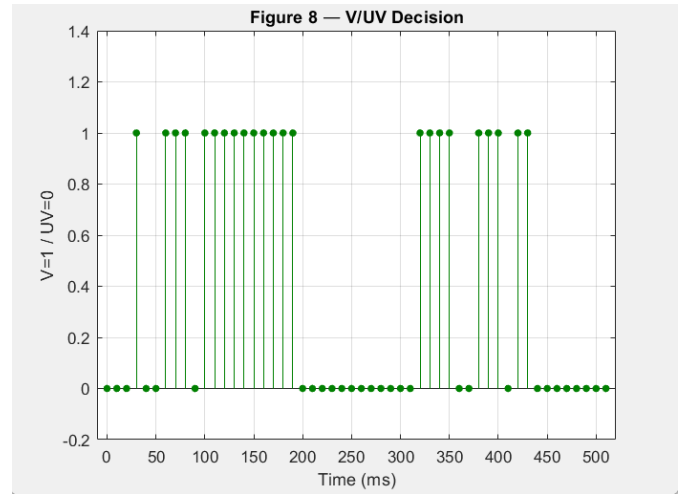


Fig. 10. Voiced (1) / unvoiced (0) decision per frame.

F. Spectral-Envelope Validation

Fig. 11 overlays, for a representative voiced frame, the smooth LPC spectral envelope on the high-resolution frame DFT. The all-pole envelope follows the formant peaks of the measured spectrum, confirming that a 10th-order model adequately represents the vocal-tract response.

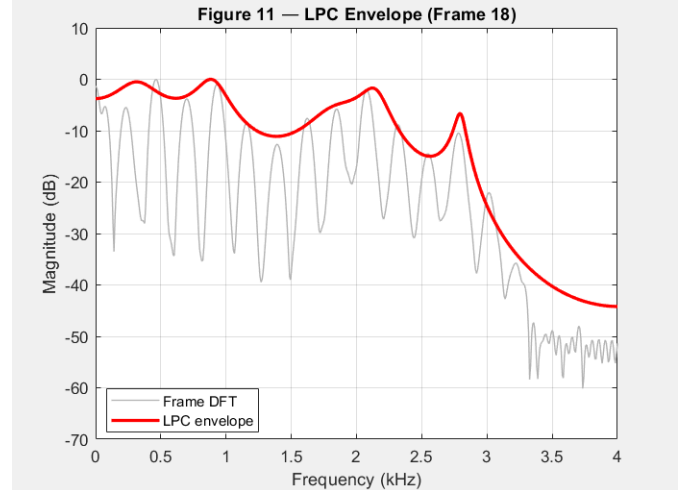


Fig. 11. LPC spectral envelope (red) compared with the frame DFT (grey) for a voiced frame.

V. PARAMETER QUANTIZATION AND BINARY REPRESENTATION

Each frame is encoded into a fixed-length 97-bit packet by uniform scalar quantization. A parameter x in a range $[x_{\min}, x_{\max}]$ is mapped to a b -bit index by

$$q = \text{round}((x - x_{\min}) / (x_{\max} - x_{\min}) \cdot (2^b - 1)). \quad (7)$$

The LSFs are quantized over $(0, \pi)$, the gain over its observed min-max range, and the pitch lag over the 20–160-sample search window. Each index is written most-significant-bit first, and the voicing flag is appended as a single bit. The per-frame bit budget is summarized in Table I.

TABLE I
Per-Frame Bit Allocation

Parameter	Bits each	Subtotal
LSF (10 coeff.)	8	80
Gain	8	8
Pitch	8	8
Voiced/unvoiced flag	1	1
Total per frame		97

A. Data-Size Comparison

Uncompressed 16-bit PCM at 8 kHz requires $8000 \times 16 = 128$ kbit/s. The parametric coder emits one 97-bit packet every 10 ms frame, so the parameter bit rate is

$$R_{LPC} = 97 \text{ bits} / 10 \text{ ms} = 9.7 \text{ kbit/s.} \quad (8)$$

Averaged over the complete set of analysis frames, this corresponds to the overall compression ratio of 13.2:1 relative to the 128 kbit/s PCM stream that is reported by the system (Fig. 20, Table II). The result is a more than thirteenfold reduction in transmitted data while preserving the perceptually important structure of the speech, which directly addresses the project requirement of comparing data size before and after the DSP-based representation.

VI. BPSK TRANSMITTER

The bitstream is mapped to antipodal BPSK symbols by $s = 2b - 1$, placing the constellation points at +1 and -1 on the in-phase axis (Fig. 12). Antipodal signaling maximizes the distance between symbols for a given energy and gives BPSK its excellent noise immunity.

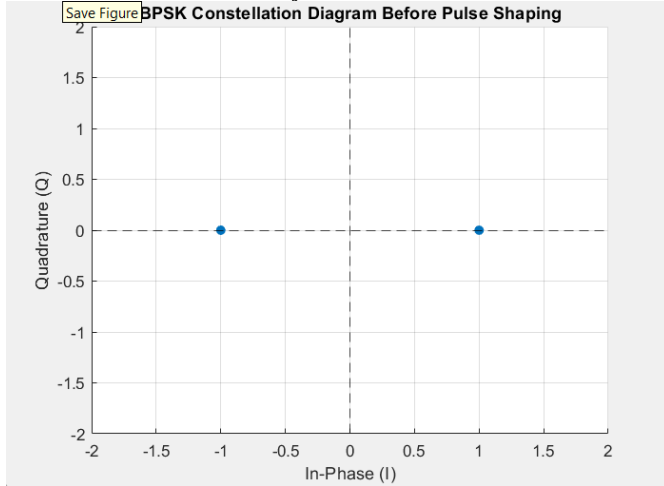


Fig. 12. Ideal BPSK constellation before pulse shaping.

To confine the transmitted spectrum and eliminate inter-symbol interference (ISI), the symbols are pulse-shaped by a square-root raised-cosine filter with roll-off $\beta = 0.5$, a span of 6 symbols, and 8 samples per symbol. A matched RRC at the receiver completes a raised-cosine response that satisfies the Nyquist ISI criterion. The occupied baseband bandwidth is approximately $(1 + \beta)R_s/2$. The shaped baseband waveform is shown in Fig. 13, and its eye diagram (Fig. 14) exhibits a wide, clean opening, indicating negligible ISI and a large timing margin. Fig. 15 overlays the discrete BPSK symbols

with the continuous RRC-shaped baseband. The waveform is finally cast to complex baseband for the USRP front-end.



Fig. 13. RRC pulse-shaped transmit baseband waveform.

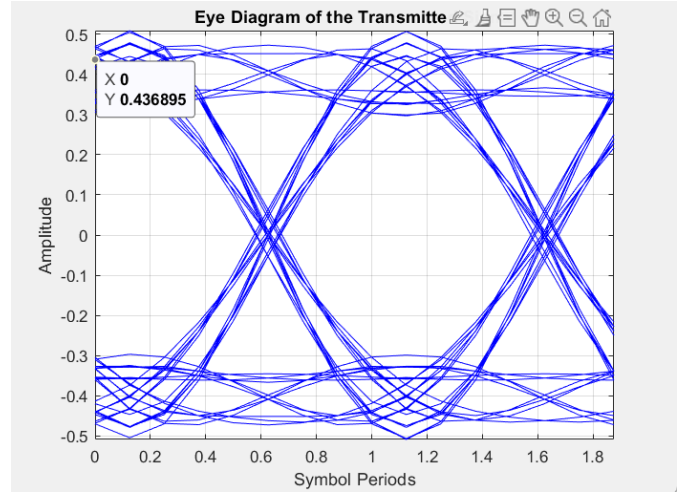


Fig. 14. Eye diagram of the transmitted signal; the wide opening indicates low ISI.

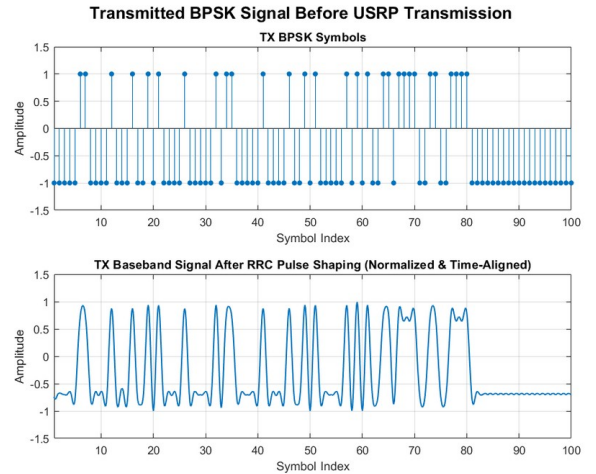


Fig. 15. Transmitted BPSK symbols (top) and the time-aligned RRC baseband signal (bottom).

VII. CHANNEL AND RECEIVER

A. AWGN Simulation

The link is first characterized in an ideal AWGN channel. With an energy-per-bit-to-noise ratio $E_b/N_0 = 10$ dB, the sample-rate SNR is

$$\text{SNR} = E_b/N_0 + 10 \cdot \log_{10}(\text{sps}) = 10 + 9.0 = 19.0 \text{ dB}. \quad (9)$$

After matched filtering, downsampling to one sample per symbol, and a simple sign decision, the recovered bits are compared with the transmitted bits. The simulated link produces zero bit errors over the full packet, against a theoretical BPSK bound of

$$\text{BER} = \frac{1}{2} \cdot \text{erfc}(\sqrt{E_b/N_0}) \approx 3.9 \times 10^{-6}, \quad (10)$$

which predicts well under one expected error in a packet of a few thousand bits — fully consistent with the observed error-free reception. Fig. 16 shows the tight symbol clusters at ± 1 , and Fig. 17 compares the transmitted and received baseband waveforms, where the AWGN appears as a low-level perturbation that does not affect the decisions.

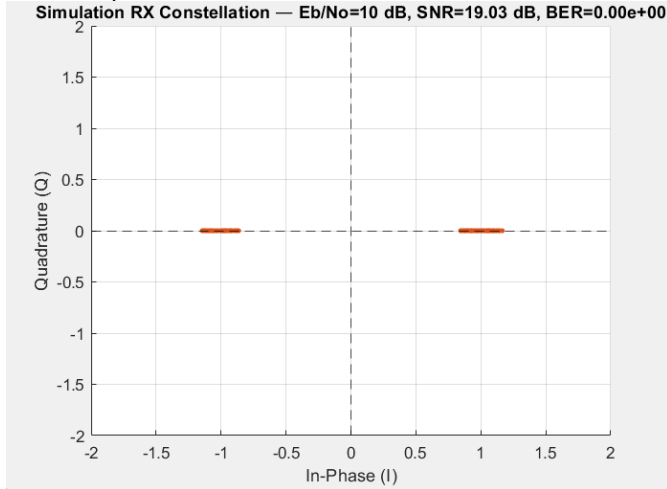


Fig. 16. Simulation RX constellation at $E_b/N_0 = 10$ dB (SNR ≈ 19.0 dB), BER = 0.

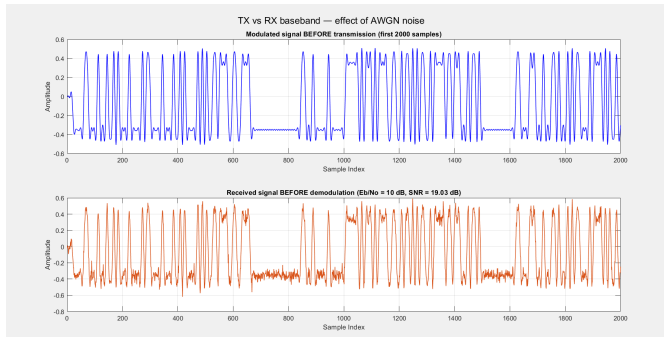


Fig. 17. Transmitted vs. received baseband under simulated AWGN ($E_b/N_0 = 10$ dB). The noise is a small perturbation of the clean waveform.

B. USRP Hardware Transmission

The same waveform is then transmitted physically through a USRP B200 radio configured at a 900 MHz carrier and 50 dB transmit/receive gain, with the complex baseband sent and captured over the air. The physical channel introduces impairments absent from the simulation: carrier-frequency offset (CFO) between the independent transmit and receive oscillators, unknown propagation delay, an arbitrary 180° carrier-phase ambiguity, automatic-gain variations, and multipath.

Synchronization is performed in three stages. First, frame timing is recovered by cross-correlating the DC-removed signal envelope of the received capture with that of the known transmit waveform; using the envelope rather than the raw signal makes the timing estimate robust to CFO. Second, after matched filtering, a block-wise (200-symbol) squared-law estimator removes the residual carrier phase by rotating each block by $\exp(-j\phi)$, where 2ϕ is the angle of the mean squared symbol. Third, the inherent 180° ambiguity of a squaring estimator is resolved by inverting all bits when the measured error count exceeds one half, which corresponds to a flipped constellation.

Receiver quality is quantified by an error-vector-magnitude SNR,

$$\text{SNR}_{\text{EVM}} = 10 \cdot \log_{10}(P_{\text{signal}}/P_{\text{error}}) = 0.7 \text{ dB}, \quad (11)$$

computed from the deviation of the equalized symbols from their ideal ± 1 positions. The hardware constellation (Fig. 18) shows the expected two clusters near ± 1 but heavily spread and rotated by the residual CFO, consistent with the low EVM-SNR. The corresponding bit-error rate is 2.7×10^{-2} . Fig. 19 collects the receiver diagnostics: the full and zoomed received waveform, the first recovered hard bits, and the block synchronization score, whose sharp peak confirms correct frame alignment.

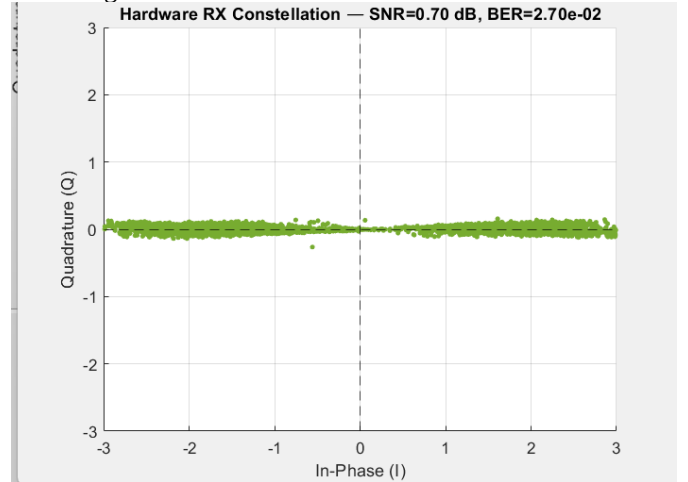


Fig. 18. Hardware (USRP) RX constellation: SNR ≈ 0.7 dB, BER = 2.7×10^{-2} .

Receiver Debug View: USRP RX Signal, Hard Bits, and Synchronization

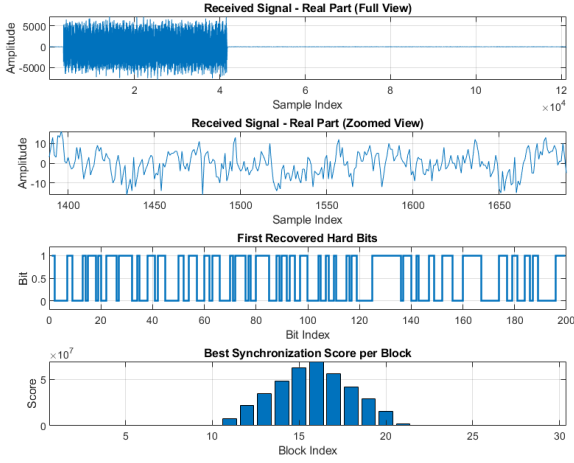


Fig. 19. Receiver debug view for the USRP link: received signal (full and zoomed), first recovered hard bits, and per-block synchronization score.

VIII. RECEIVER-SIDE SPEECH SYNTHESIS

The recovered bits are de-serialized into per-frame indices and de-quantized to LSFs, gain, pitch, and voicing. The LSFs are converted back to LPC coefficients, and any synthesis-filter root that has drifted outside the unit circle (because of bit errors) is reflected back inside to guarantee stability. The excitation is generated according to the voicing flag: a periodic impulse train at the decoded pitch lag for voiced frames, and white Gaussian noise for unvoiced frames. Each excitation is scaled by the decoded gain and passed through the all-pole synthesis filter. The synthesized frames are recombined by Hamming-windowed overlap-add with window-sum normalization, and the 8 kHz result is upsampled to 48 kHz for playback. The reconstructed speech is intelligible, confirming that the end-to-end chain — from corrupted input to over-the-air reconstruction — functions as intended.

IX. RESULTS AND PERFORMANCE EVALUATION

The complete system is summarized quantitatively in Figs. 20–22 and Table II. Fig. 20 is a single-page dashboard combining the BER and SNR comparisons, the compression ratio, representative transmit and receive waveforms, and both constellations. Fig. 21 isolates the headline BER and SNR comparisons between the simulation and the hardware link, and Fig. 22 places the two operating points on the theoretical BPSK BER-versus-SNR curve.

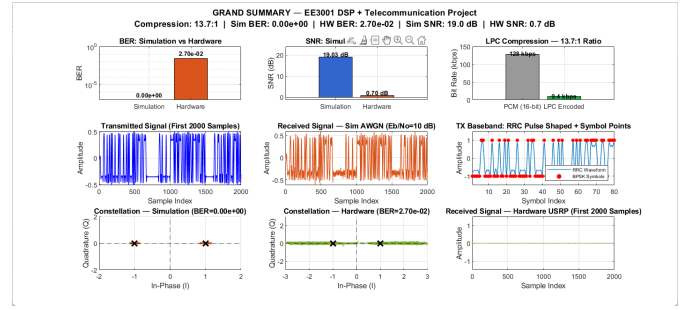


Fig. 20. Grand summary dashboard of the complete DSP + telecommunication system.

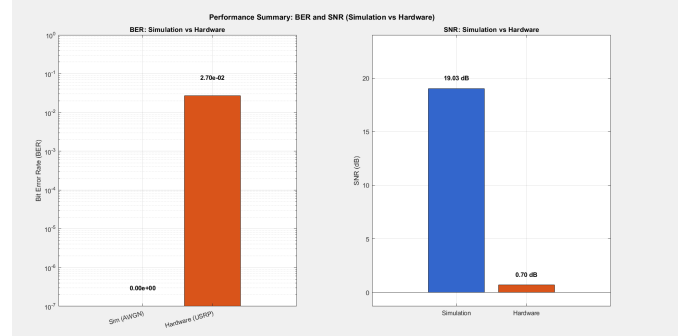


Fig. 21. Performance summary: BER (left) and SNR (right) for the simulated AWGN link vs. the USRP hardware link.

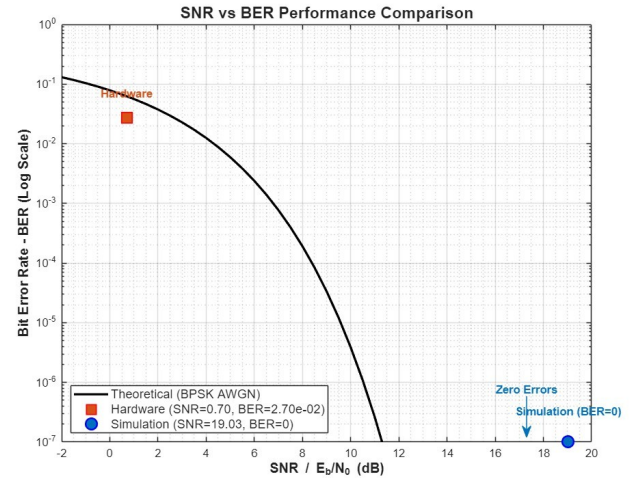


Fig. 22. BER vs. SNR. The theoretical BPSK-in-AWGN curve, the error-free simulation point, and the hardware operating point are shown.

TABLE II
Simulation vs. Hardware Performance

Metric	Simulation	Hardware
Channel	AWGN	Over-the-air
SNR (dB)	19.0	0.7
BER	0	2.7×10^{-2}
Carrier	—	900 MHz
Compression	13.2 : 1	13.2 : 1

The data-size comparison confirms the 13.2:1 reduction of Section V. In simulation the link is effectively error-free, matching the theoretical expectation at 19.0 dB SNR. The

hardware link, operating at only 0.7 dB SNR, exhibits a measured BER of 2.7×10^{-2} .

X. DISCUSSION

The roughly 18 dB SNR gap between simulation and hardware, and the corresponding rise in BER from zero to a few percent, is the central experimental observation. Several physical effects explain it. Over-the-air propagation incurs path loss and picks up environmental interference that the simulation never models. A residual carrier-frequency offset rotates the constellation between synchronization blocks and is only partially removed by the block-wise tracker, broadening the clusters in Fig. 18. The squaring-based phase estimator and the envelope-based timing recovery, while robust, are coarse compared with a dedicated pilot-aided receiver. Finally, automatic gain control and half-duplex loopback timing introduce amplitude and alignment variability.

The effect of these bit errors on speech quality depends on which bit is hit, and the parametric structure provides graceful degradation. The ordered, bounded nature of LSFs means that isolated bit errors typically cause bounded spectral distortion rather than filter instability, and the root-reflection safeguard prevents divergence. Errors in the gain produce loudness fluctuations, errors in the pitch lag cause short pitch “warble,” and a flipped voicing bit briefly substitutes noise for periodic excitation or vice versa, heard as buzz or hiss. Because each 10 ms frame is independent, an error is confined to that frame and the speech remains intelligible at a 2.7% BER, though audibly degraded relative to the simulation.

Several improvements follow directly. Adding forward error-correction coding (the project’s extra-credit option) would protect the most sensitive bits and sharply reduce the effective BER. A pilot-assisted or decision-directed carrier and timing loop would tighten synchronization, and increasing transmit gain or reducing the antenna separation would raise the physical SNR. On the source-coding side, vector quantization of the LSFs and differential coding of slowly varying parameters could lower the bit rate further below 9.7 kbit/s with little additional distortion.

XI. CONCLUSION

A complete wireless speech communication system combining LPC/LSF source coding with a BPSK USRP physical layer was designed, implemented, and evaluated. Out-of-band corruption was removed by zero-phase filtering before decimation to 8 kHz; a 10th-order LPC/LSF analyzer with autocorrelation pitch estimation and a three-feature voicing classifier produced a 97-bit/frame representation that compresses the data by 13.2:1. The bitstream was BPSK-modulated, RRC-shaped, and transmitted both through an ideal AWGN simulation (19.0 dB SNR, BER = 0) and physically over a USRP B200 link at 900 MHz (0.7 dB SNR, BER = 2.7×10^{-2}), and intelligible speech was reconstructed at the receiver in both cases. The results quantify both the substantial source-coding gain and the realistic penalty imposed by a physical radio channel, and they identify error-correction coding and improved synchronization as the most promising next steps.

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